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Srihari Maruthachalam

Chief Technology Officer and hands-on Machine Learning leader with over a decade across data science, research, and engineering. Currently driving agentic AI products end-to-end at Qumin (custom LLM orchestration, computer vision, and autonomous testing), after shipping real-time edge AI at scale in automotive safety and healthcare. A proven record of turning deep ML expertise into products, teams, and measurable impact.

Professional Experience

Mar 2026 - Present | **Chief Technology Officer**, *Qumin.ai*, Bengaluru

Lead the technology organization for two agentic AI products, heading a team of 4 engineers and owning architecture, ML strategy, infrastructure, and delivery end-to-end.

Drive an AI-powered no-code QA platform (nocode.qumin.ai) that converts plain-English test cases into autonomous end-to-end tests. Designed a custom agent orchestration engine (no off-the-shelf frameworks) where an LLM plans and executes user journeys through a Playwright execution layer, while a vision + OCR pipeline interprets rendered UI like a human, making tests resilient to DOM and layout changes. Engineered the entire agent loop to run reliably on Gemini 3.1 Flash Lite, achieving frontier-grade test accuracy at a fraction of typical inference cost, the platform's core technical moat. Deployed on AWS; additionally architected the Android QA platform extending agentic testing to mobile.
• LLM Agents • Custom Orchestration • Computer Vision • OCR • Playwright • Gemini • AWS

Expanded the company into chat-first sales prospecting (qumin.ai): a natural-language prompt is converted into an explicit ICP, after which autonomous research agents mine the open web for buying signals (funding, hiring, tech adoption), enrich and cross-verify contacts with confidence scores, and stream source-backed evidence into a live human-in-the-loop workspace, built on the same custom orchestration stack.
• Agentic Workflows • Web-scale Data Extraction • Entity Resolution • Ranking & Scoring

Jul 2025 - Feb 2026 | **Career Break**, *Parental leave: birth of my child*

Jun 2022 - Jun 2025 | **Staff Data Scientist**, *Netradyne Technology India Pvt Ltd*, Bengaluru

Contributed to a safety initiative by developing XGBoost (XGB) and LightGBM (LGBM) models to classify potential collisions using an inertial and visual data stream. The models, trained on signal processing features, use a trigger point from the inertial jerk. Achieved a recall over 96% and a precision over 95% for potential collision class on the edge. Further enhanced by a Deep Neural Network (DNN) model on the cloud, increasing the recall by an additional 2%.
• Edge computing • Multiclass Classification • Deep Neural Network • PyTorch • XGB • LightGBM

Implemented a fuel-saving initiative by developing a neural network (NN) model that utilized inertial and GPS features to identify vehicle idling for every minute, despite unreliable GPS data. When prolonged idling is detected, audio feedback is played in real-time (RT) to prompt the driver to switch off the engine. This model achieved a 76.3% F1 score in production and has been estimated to contribute to the environment by reducing ∼150 tons of CO_2 emissions monthly.
• Neural Network • Binary Classification

Pioneered a safety initiative by developing a threshold-based method to identify hard braking instances, an indicator of reckless driving, using speed and inertial data streams. By leveraging existing visual models, this approach also determines the causes of hard braking in RT. This critical information forms the basis of a coaching program, which has reduced driver distraction by 67%, enhancing overall road safety.
• Real-time Analysis

Apr 2020 - Jun 2022 | **Data Science Associate Consultant**, *ZS Associates*, Bengaluru

Developed a BERT transformer model to diagnose rare diseases, addressing challenges in healthcare due to misdiagnosis and delayed treatment. The model was trained on administrative claims datasets comprising patient-level claims data, diagnosis codes, procedures, and medications. This approach achieved a test AUPRC of 96.8%, outperforming the baseline XGB and LSTM models, which scored 79% and 80.1%, respectively.
• Transformer • Python • LSTM • PU Learning

Developed a genetic algorithm for automated feature selection, evolving feature sets via fitness-based selection, crossover, and mutation, and injecting (non-)linear transformations of high-importance features each generation to improve downstream model performance.
• XGB • Feature engineering

Contributed to a healthcare initiative by developing an XGB model to predict if oncology patients would progress to the next stage within six months, using insurance claims data as features. Further enhanced this predictive model by building a Convolutional NN (CNN) version that processed a matrix of claims and prescriptions over time. This model outperformed the initial XGB model by an absolute 13.7% in AUPRC.
• CNN • PyTorch

Built a regionalized OLS marketing-mix model identifying the most effective sales channels for a pharmaceutical drug, achieving R^2 of 0.76. • OLS • Residual Analysis

Jul 2019 - Feb 2020 **Project Associate**, *Indian Institute of Technology, Madras*, Chennai
Developed an Android app for Automated Speech Recognition, converting Indic speech into text, in collaboration with Electrical Engineering and Computer Science departments.
Led the creation of an Android app translating eye blinks into speech using a single-channel remote EEG, revolutionizing communication for individuals with speech and movement limitations.
Formulated the SEAT project, a graph matching algorithm mapping student preferences to course enrollment, optimizing the process by considering class capacity and prerequisites.
• Java • Python • Git • ASR • Neural Network

Dec 2016 - Jun 2019 **Research Assistant**, *Indian Institute of Technology, Madras*, Chennai
MHRD-funded Half-Time Research Assistantship. Teaching Assistant for Introduction to Programming and Pattern Recognition: course curation, question papers, and evaluation.
• NumPy • PyTorch • MATLAB • Git • LaTeX

Aug 2014 - Jan 2017 **Programmer Analyst**, *Cognizant Technology Solutions India Pvt Ltd.*, Chennai
Full-stack developer: built a workflow management system for a US insurer, a web publishing application, and a job monitoring system, saving ~50 human hours monthly.
• ASP.Net MVC • jQuery • MS SQL Server

Educational Qualifications

2016 - 2020 **Indian Institute of Technology**, *Madras*, MS by Research in Computer Science and Engineering, 8.0/10

2010 - 2014 **Anna University**, *Chennai*, B.Tech. Information Technology, 8.2/10

Publications

EMBC 2019 **Srihari Maruthachalam**, Mari Ganesh Kumar, and Hema A Murthy, *Time Warping Solutions for Classifying Artifacts in EEG*, 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) 2019, Germany

Interspeech 2018 **Srihari Maruthachalam**, Sidharth Aggarwal, Mari Ganesh Kumar, Mriganka Sur, and Hema A Murthy, *Brain-Computer Interface using Electroencephalogram Signatures of Eye Blinks*, Proc. Interspeech 2018, 1059-1060, Hyderabad, India

Scholastic Achievements

GATE 2016 Top 1.5% among all the candidates

Selected Awards

Dec 2024 Aspire Leadership Programme, Netradyne

Oct 2024 Netradyne Information Security Ambassador (NISA), Netradyne

Dec 2023 Dream Team Award, Netradyne